

Qatar-4b

R-2633

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_191004 · created 2026-08-29T00:37:34+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458760.7138 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.095 +/- 0.044
Transit depth (Rp/Rs)^2	0.91 +/- 0.83 [%]
Orbital Inclination (inc)	85.84 +/- 2.14
Airmass coefficient 1 (a1)	0.916 +/- 0.066
Airmass coefficient 2 (a2)	0.079 +/- 0.059
Scatter in the residuals of the lightcurve fit is	7.25 %
Best Comparison Star	#1 - [369, 134]
Optimal Aperture	3.14
Optimal Annulus	11.82
Transit Duration (day)	0.074 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.095 $\pm$ 0.044 vs 0.138 (deviation 0.7 σ)
Duration fit vs published	0.074 d vs 0.0892 d (deviation 0.63 σ)
Mid-time O-C	3.2 min
Depth SNR	1.5
Residual scatter	7.25 %

Light curve

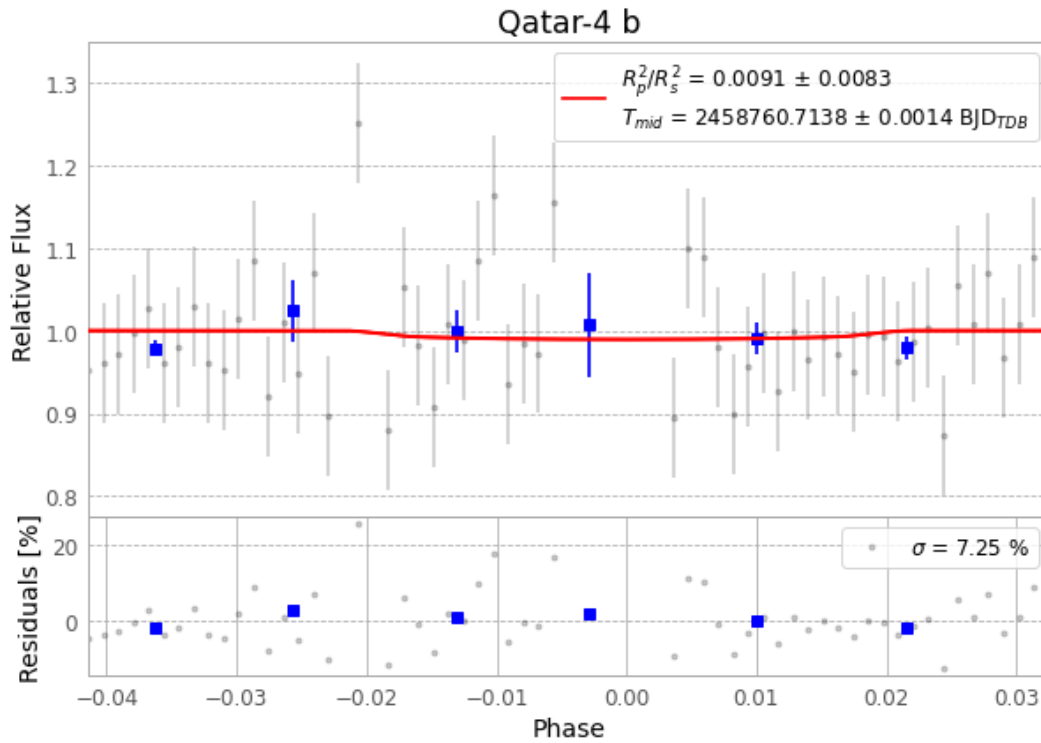


Figure 1 — FinalLightCurve_Qatar-4 b_2019-10-03.png (SHA-256 e695a28bdb8dd9cb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [411, 153], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 37d04285482514d1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), Qatar-4191004031212.FITS → Qatar-4191004062412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-191004.log (68315030ffd7...), FinalLightCurve_Qatar-4 b_2019-10-03.png (e695a28bdb8d...), FinalLightCurve_Qatar-4 b_2019-10-03.csv (2e8e574a2d56...)

Compute resources

Wall time 128.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 00:37Z.